

Files, coordinates and reference versions

Are these two files describing the same stretch of DNA?

By the end, you should be able to...

Recognize the purpose of common biological file formats.

Convert a positive-length interval between BED and one-based inclusive coordinates.

Report an unknown reference build rather than guess.

Reconnect with the biology

A genome is a sequence; an annotation adds named features on a particular sequence version.

Think silently -> explain to a neighbour -> check the measurement.

Six file families; six jobs

File	Primary question
FASTA / FASTQ	What sequence, with what quality?
SAM / BAM	Where did a read align?
BED / GTF / GFF	Where is an interval or feature?
VCF	What differs from the reference?

FASTA: sequence. FASTQ: sequence plus per-base quality.

SAM/BAM: read alignments. BED: genomic intervals.

GTF/GFF: annotated features. VCF: sequence variants.

FASTA does not guarantee a reference build

```
>toy_contig  
ACGTACGTAA
```

A header can give an identifier, but it may omit assembly metadata.

Sequence names such as chr1 do not uniquely identify an assembly.

Write unknown and request metadata when evidence is absent.

Two coordinate rulers

Same bases. Different ruler.

1	2	3	4	5	6	7	8	9	10
A	C	G	T	A	C	G	T	A	A
0	1	2	3	4	5	6	7	8	9

1-based positions above / 0-based indices below

One-based inclusive: first base is position 1; both endpoints count.

BED: start is zero-based; the end is excluded.

One-based [3, 7] and BED [2, 7) describe the same five bases.

Keep the interval length invariant

BED [2, 7) $\leftrightarrow$ inclusive [3, 7]

State the units and the assumptions.

Inclusive length = end - start + 1.

BED length = end - start.

To convert a positive-length BED interval, add one to start only.

Why adjacency is not overlap

Same bases. Different ruler.

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A	C	G	T	A	C	G	T	A	A
0	1	2	3	4	5	6	7	8	9

1-based positions above / 0-based indices below

BED [0, 5) contains indices 0, 1, 2, 3, 4.

BED [5, 8) contains indices 5, 6, 7.

These intervals touch but share no base.

Reference version versus chromosome name

Naming convention

Reference assembly

GRCh37 and GRCh38 are different human assemblies.

chr1 versus 1 can be a naming difference, not an assembly conversion.

Renaming a chromosome does not lift coordinates between assemblies.

Read text and binary formats carefully

SAM text POS uses one-based positions.

BAM stores coordinates internally using a different convention.

A software API may expose zero-based values even when the input file is one-based.

GENOMES: your reusable audit card

1 G: reference + annotation; E: environment + seed; N: dimensions.

2 O: coordinates; M: multiplicity + independent unit.

3 E: stable entities; S: sources. Use justified N/A where a check does not apply.

Worked example: recover the same bases

Toy FASTA sequence ACGTACGTAA; select one-based positions 3 through 7.

Predict the result and one limitation before calculating.

Work through the reasoning

Inclusive length: $7 - 3 + 1 = 5$.

Equivalent BED interval: start 2, end 7.

Python `seq[2:7]` returns GTACG, five bases.

Pause: make a defensible claim

Convert BED [4, 9) to one-based inclusive coordinates. Can a FASTA header reading only >chr1 prove GRCh38?

First answer alone. Then compare the evidence, not just the conclusion.

A defensible response

The interval is $[5, 9]$, length 5. The header alone does not prove the assembly.

Today's practical

W02: inspect toy FASTA, BED, GTF and VCF records; fix a start-coordinate error and record reference provenance.

Minutes	Action
0-10	Predict + set up
10-30	Run + inspect + check
30-40	Explain + compare
40-45	Save + submit / exit

Five terms to use precisely

- 1 reference
- 2 annotation
- 3 interval
- 4 inclusive
- 5 half-open

Exit ticket

What did we actually measure or calculate?

What is one assumption or limitation of that result?

What check would you perform before sharing it?

Go further only after the core is clear

Optional: inspect how a VCF deletion includes an anchor base; no indel normalization is examined.

Not required for the core grade. No advanced model training is expected.

Read the evidence, not only the slide

S03 SAM/BAM, VCF and BED specifications, maintained by the HTS community.

S04 UCSC Genome Browser: data format FAQ.

S05 Python 3 tutorial: introduction and control flow.